

Assignment 3

10/7/2026

22 Points Possible

▼ Details

Curtis problems 4.3, 4.4, 4.5, 4.6, 4.7

For problems 4.3, 4.4, and 4.5, also compute the equinoctial orbital elements a , h (this is not angular momentum), k , p (this is not semilatus rectum), q , λ .

Where classical orbital elements are requested, use standard orbital elements, with semimajor axis a , not angular momentum magnitude h , per HW2.

Find the equinoctial elements for the following orbits:

- Position and velocity 3698.3455, -34232.4473, 0., 2.1482, -1.4944, 0. (km, km/s)
- Position and velocity -21082.0848, 36515.242, 0., -2.6627, -1.5373, 0. (km, km/s)
- Position and velocity -15843.4562, -2247.7766, 21200.4627, 1.4544, -3.5188, 0.7138 (km, km/s)